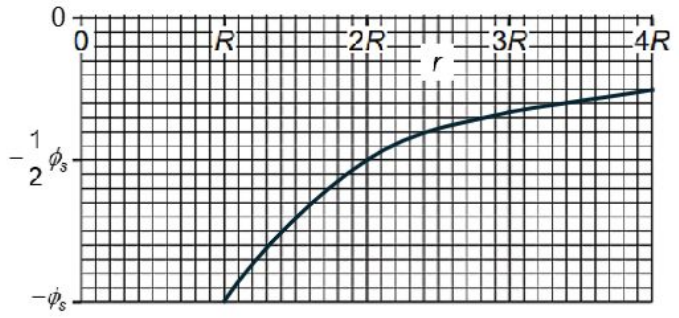
3	(a)(i)	curve with continuously decreasing magnitude and with gradient of continuously decreasing magnitude, starting at (R, $-\phi_s$) line passing through ($2R$, $-½\phi_s$) and ($4R$, $-¼\phi_s$)  	B1
	(a)(ii)	Gravitational potential at infinity is taken to be zero. OR An external force has to be applied in opposite direction to gravitational force. (opposite to the displacement) Work done to bring a mass from infinity to the point in the field is negative.	B1 B1
	(b)	$\phi = -\frac{GM}{r}$ GM = gradient of the graph $M = \frac{1.76\times10^{8}}{3.0\times10^{-8}\times6.67\times10^{-11}}$ $M = 8.8\times10^{25}$ kg Accept if student just use one point on the graph.	 M1 A0
	(c)(i)	Gravitational force provides the centripetal force for the circular motion. $F_G = F_C$ $\frac{GMm}{r^2} = mr(\frac{2\pi}{T})^2$ $\frac{(6.67\times10^{-11})(8.8\times10^{25})}{r^2} = r(\frac{2\pi}{0.72\times24\times60\times60})^2$ $r = 8.3\times10^{7}$ m If student did not convert the 0.72 Earth days to seconds, deduct 1 mark for this question.	B1 C1 A1
	(c)(ii)	Using principle of conservation of energy $GPE_i + KE_i$ + Additional energy = $GPE_f + KE_f$ $GPE_i + KE_i$ + Additional energy = 0	 C1

		$-\frac{GMm}{r}+(\frac{GMm}{2r})$ + Additional energy = 0 Additional energy = $\frac{GMm}{2r}$ = $= \frac{(6.67\times10^{-11})(8.8\times10^{25})(1200)}{2(8.3\times10^{7})}$ $= 4.24\times10^{10}$ J	**C1** **A1**